MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4SECECT201 PCB Design & 3D Printing

Duration: 1½ hours **Maximum Marks: 50**

**Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions, Each question carries 1 mark

1	Which electronic component allows current to flow only in one direction? A) Resistor B) Capacitor C) Diode D) Inductor	U	1
2	3D printing is best described as which type of manufacturing? A) Subtractive B) Formative C) Additive D) Casting	U	2
3	Which PCB layer contains component labels and markings? A) Copper layer B) Substrate C) Solder mask D) Silkscreen	U	1
4	Which 3D printing technique uses liquid resin cured by a laser? A) FDM B) SLA C) SLS D) CNC	U	2
5	Which of the following is a passive electronic component? A) Transistor B) IC C) Capacitor D) Op-amp	U	1
6	Which file format is commonly used for 3D printing? A) PDF B) JPG C) STL D) DOC	U	2
7	What does LDR stand for? A) Light Density Resistor B) Low Data Resistor C) Light Dependent Resistor D) Laser Driven Resistor	U	1
8	Which process converts a 3D model into G-code? A) Designing B) Printing C) Slicing D) Rendering	U	2
9	Who developed the first printed circuit board (PCB)? A) Nikola Tesla B) Thomas Edison C) Paul Eisler D) James Watt	U	1
10	Which 3D printing method does not require support structures? A) FDM B) SLA C) SLS D) DLP	U	2

11	Which PCB type has copper on both sides? A) Single-sided PCB B) Double-sided PCB C) Multilayer PCB D) Flex PCB	U	1
12	Which CAD software is web-based and beginner friendly? A) SolidWorks B) Blender C) Fusion 360 D) TinkerCad	U	2
13	Design for 3D printing mainly focuses on: A) Color and texture B) Printability and functionality C) File size D) Printer brand	U	2
14	Which 3D printing material is biodegradable and easy to print? A) ABS B) PETG C) PLA D) Nylon	U	2
15	Which PCB process removes unwanted copper? A) Drilling B) Etching C) Plating D) Printing	U	1
16	Which software is mainly used for professional parametric modeling? A) TinkerCad B) Fusion 360 C) Paint 3D D) SketchUp	U	2
17	Which electronic component stores electrical energy? A) Resistor B) Capacitor C) Diode D) Transistor	U	1
18	What is the main disadvantage of FDM printing? A) High-cost B) Limited size C) Visible layer lines D) No material options	U	2
19	Which PCB layer prevents oxidation and short circuits? A) Silkscreen B) Substrate C) Copper layer D) Solder mask	U	1
20	Which CAD tool is commonly used for artistic 3D models and animation? A) Fusion 360 B) SolidWorks C) Blender D) TinkerCad	U	2
21	Which component is used to regulate voltage? A) Transformer B) Voltage regulator C) Resistor D) Relay	U	1
22	Which 3D printing method uses powdered material fused by laser? A) FDM B) SLA C) SLS D) DLP	U	2
23	Through-Hole Technology requires components to be: A) Glued on PCB B) Mounted on surface C) Inserted into holes D) Printed directly	U	1
24	Which stage improves surface finish after printing? A) Slicing B) Designing C) Post-processing D) G-code creation	U	2

25	Which PCB technology allows components on both sides with no holes? A) THT B) SMT C) DIP D) SIP	U	1
26	Which PCB file format is used for manufacturing? A) PDF B) DOCX C) JPG D) Gerber	U	1
27	Which feature allows editing dimensions later in CAD? A) Rendering B) Sketching C) Parametric modeling D) Sculpting	U	2
28	Which inductor is mainly used for noise suppression? A) Air-core B) Wire-wound C) Ferrite bead D) Toroidal	U	1
29	Which electronic component amplifies signals? A) Capacitor B) Resistor C) Transistor D) Inductor	U	1
30	Which step involves loading filament into a printer? A) Designing B) Slicing C) Printing D) Post-processing	U	2
31	Which PCB material is commonly used as substrate? A) Plastic B) Copper C) Fiberglass D) Aluminum foil	U	1
32	Which 3D printing issue causes thin unwanted strings? A) Warping B) Layer shifting C) Stringing D) Cracking	U	2
33	Which CAD software is considered industry standard for engineering? A) Blender B) TinkerCad C) SolidWorks D) SketchUp	U	2
34	Which PCB design check verifies spacing and width rules? A) ERC B) Simulation C) DRC D) Routing	U	1
35	Which 3D printing method provides the smoothest surface finish? A) FDM B) SLA C) SLS D) Binder jetting	U	2

1 * 35 = 35

Part B

Short Answer Questions

Answer any **3** Questions, Each question carries 5 marks

36	Discuss a case study where integration failure or partial success occurred in a real-world project. Analyze the reasons for failure, the lessons learned, and the improvements suggested for future integration.	An	3
37	Explain the step-by-step process of designing and fabricating a PCB for a prototype.	E	4
38	Analyze a case study of hardware–software integration in an embedded or industrial system. Identify at least three major challenges encountered during integration and explain how they were resolved.	An	3
39	Describe the steps involved in designing and 3D printing an enclosure for an electronic prototype.	E	4
40	Explain a real-world case study where multiple technologies were integrated into a single system. Discuss the objective of integration, the challenges faced, and the solutions implemented.	An	3

3 * 5 = 15